

Video Game Programming: Game Testing and Release

L21

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Version naming

- It's important to differentiate between a work in progress and a release candidate
- Naming indicates in which part of the lifecycle the game is
- There are different schemes for file naming, e.g., Alpha candidate 1.0, Beta 0.4, Release candidate 0.1
- It is important to name subsequent releases accordingly
- Naming can have different significance and purposes in game development, technical support, and marketing

Development milestones

- In the game lifecycle there are some fundamental milestones
- Most game studios use the following:
 - Prototype (pre-Alpha), i.e., the rough draft
 - Alpha, i.e., a working project that is missing significant features
 - Beta, i.e., a feature-complete version, but with bugs
 - Release candidate, i.e., a version which is candidate for release, which may still have some last bugs
 - Gold, i.e., a release ready for publishing
- While prototype and alpha version are internal, beta, release candidate and gold ones need to be distributed

Quality assurance

- **Quality assurance (QA)** is a crucial step in game development
- **Quality control** is the process of finding bugs and, crucially, sending them back to the designers and the engineers to fix
- **QA team** tests the game at functional and non-functional levels:
 - Acceptance testing: alpha, beta
 - Performance: load and stress testing
 - Usability
 - Security
 - Portability
 - ...

QA testing

- QA testing is the process of certifying that a game is ready for public consumption
- QA testers play games trying to break them, to make sure that players don't break them accidentally (or intentionally)
- In a 3D video game, some examples of QA testing include:
 - Wall testing, where the tester tries to run into every wall
 - Input testing, where the tester tries to give the game confusing or unexpected inputs

Types of bugs

- Bugs are generally categorized by:
 - How they appear
 - How serious they are
- **Showstopper** bugs prevent the game from running and must be fixed as soon as possible
- Other bugs are minor (e.g., they occur only for rare input combinations) and thus have lower severity and priority
- **Reproducibility** is a description of the steps that it takes to encounter the bug, which greatly helps fixing

Test plan

- Writing a test plan for a game means that:
 - the designer identifies potential problems / buggy areas
 - offers suggestions on how to find those bugs
- A test plan document can include test cases / scenarios
- Suggested exercise:
 - Write a test plan for a game
 - Hand the plan off to a colleague to test
 - Make tests following a colleague's plan

Test documentation and bug tracking

- It is important to document all testing stages
- Besides the test plan, test documents include a journal / log to keep track of bugs (open and closed)
- Complex game projects can benefit of a bug / issue tracking software (e.g., Bugzilla)
- In the software / log, bugs can have an ID, reproducibility information, severity, priority, status, comments and votes...

Troubleshooting strategies

- Troubleshooting strategies aim at identifying and resolving bugs
- To troubleshoot a bug, developers should:
 - reproduce it
 - identify it in code
 - and then fix it
- Creating cause-and-effect diagrams to account for possible causes can be helpful
- Reproducing and locating bugs is not easy and can be frustrating
- Game engines and visual editors offer debugging tools

Unity debugging hints

- Look at the line indicated by the error message
- Set breakpoints
- Analyze call stack
- Print out values on console
- Adopt proper formatting techniques
- Check code structure
- Check for misspelled words
- Check parameters and return values
- Look for incorrect values in loops and conditions
- Fix one error at a time and retest

QA beyond bugs

- Bugs and performance are usually considered the most relevant aspects of commercial game QA
- Indeed, several other aspects are increasingly considered both by academics and companies, before and after game release
- Most of these additional aspects concern user experience (UX), including likability, flow, engagement
- In addition, for serious games, it is fundamental to test their effectiveness in terms of knowledge gain, attitude change,...

Questionnaires

- Questionnaires are a well-known instrument in psychology, education, and human-computer interaction
- They can be administered
 - before and / or after releasing the game
 - before and / or after playing
 - in a lab setting and / or in the wild
 - using paper & pencil and / or electronic surveys
- The literature provides questionnaires for lots of different aspects
- In some cases, specific questionnaires should be designed

Game engagement and experience

- GEQ is an acronym used for 2 different questionnaires:
 - Game Engagement Questionnaire can be used to assess engagement
 - Game Experience Questionnaire can be used to assess players' experience
- GEngQ includes 19 statements and participants are asked to rate each of them on a 3-point scale (No, Sort of, Yes)
- GExpQ core module includes 33 statements concerning feelings while playing and participants rate them on a 5-point scale
- GExpQ has other 3 modules with additional items

Experience inventory

- The Player Experience Inventory (PXI) measures player experience
 - at Functional Consequences level, i.e., the immediate experiences as a direct result of game design choices
 - at Psychosocial Consequences level, i.e., second-order emotional experiences
- It has 30 items rated on a 7-point Likert scale

FUNCTIONAL consequences

Ease of control

Progress feedback

Audiovisual appeal

Goals and rules

Challenge

PSYCHOSOCIAL consequences

Mastery

Curiosity

Immersion

Autonomy

Meaning

Game flow

- The EGameFlow questionnaire assesses 8 constructs concerning games:
 - concentration, i.e., how much the participant can remain concentrated on the game
 - goal clarity, i.e., how much the goals of the game are clear
 - feedback, i.e., quality of perceived feedback provided by the game
 - challenge, i.e., how appropriate is the level of challenge and its increase
 - autonomy, i.e., sense of control on the game perceived by the participant
 - immersion, i.e., how much the participant feels immersed in the game
 - social interaction, i.e., how much the game becomes a mean for social interaction
 - knowledge improvement, how much players feel that the game improved their knowledge
- The questionnaire asks participants to rate 42 items on a 7-point Likert scale and averages the items for each construct to form reliable scales
- The first 6 constructs are taken from GameFlow questionnaire and are connected to Csikszentmihalyi's concept of flow experience

Presence

- Presence can be measured with the Igroup Presence Questionnaire
- IPQ is a self-report scale with 14 items: a general item related to the sense of “being there” and three subscales (spatial presence, involvement, experienced realism)
- Each item can be rated on a 7-point scale
- It is available at <http://www.igroup.org/pq/ipq/index.php>
- There are also other presence questionnaires such as Witmer, Jerome & Singer’s 29-item PQ

Usability

- The well-known System Usability Scale (SUS) can be used to measure usability
- SUS includes 10 statements and participants rate their level of agreement on a 5-point Likert scale
- Studies report an overall SUS score in the 0-100 range
- In some studies, a SUS score above a 68 would be considered above average

Usefulness and satisfaction

- The well-known USE Questionnaire can be used to measure ease of use and learning as well as usefulness and satisfaction
- It includes 30 statement:
 - 8 about usefulness
 - 11 about ease of use
 - 4 about ease of learning
 - 7 about satisfaction
- Participants rate their level of agreement on a 7-point Likert scale

Comfort

- The Device Assessment Questionnaire (DAQ) can be used to measure comfort
- It was designed to compare pointing devices for computers
- It includes 13 items concerning required force, smoothness, mental and physical effort, difficulty to be accurate, slowness, fatigue of different body parts, general comfort and ease of use
- Each item can be rated on a 5-point scale

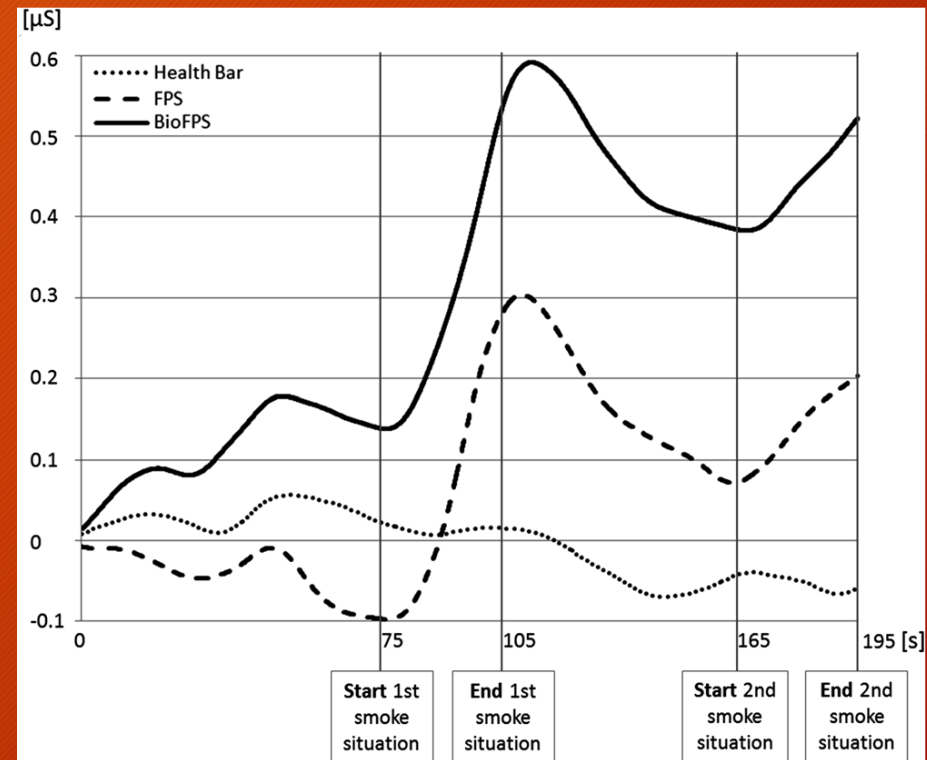
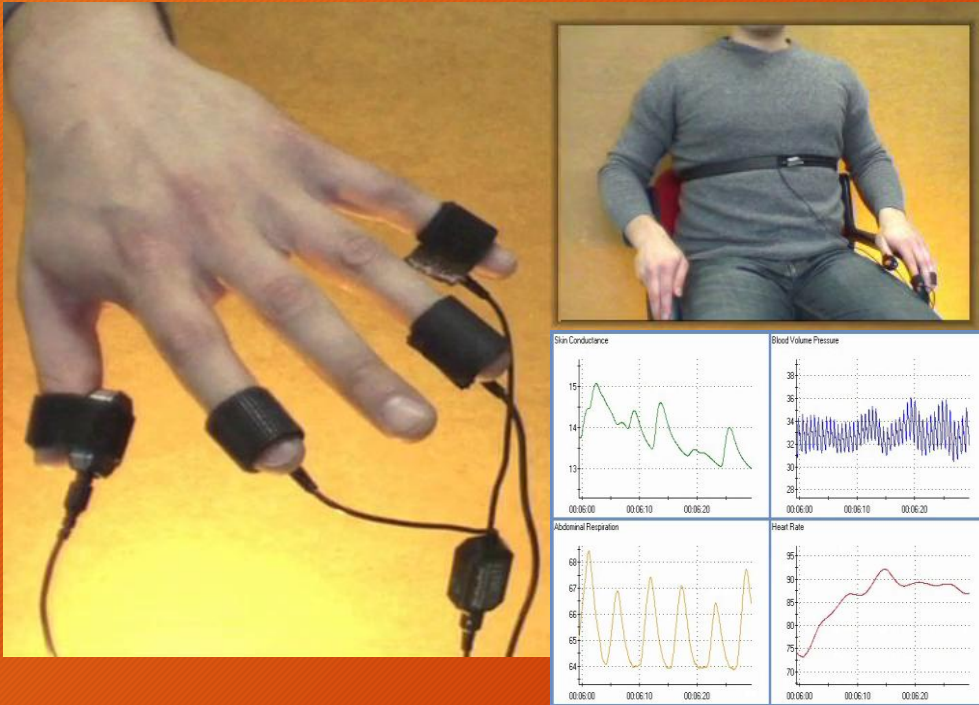
Sickness

- To assess sickness, participants rate how much they are affected by different symptoms after completing a game task (post-test)
- Usually, experimenters ask participants to rate the same symptoms also before playing (pre-test baseline)
- An increase in sickness is obtain by subtraction
- Simulator Sickness Questionnaire (SSQ) assess different symptoms such as general discomfort, headache, eyestrain, nausea,...
- SSQ asks participants to rate each symptom at the time of questionnaire completion on a 4-point scale (none, slight, moderate, severe)
- Some researchers suggest not using the (whole) questionnaire to minimize the negative influence of pre-test compilation

Physiological parameters

- Some physiological parameters can be useful to evaluate games
- E.g., level of physiological arousal elicited by a game can be measured by recording electrodermal activity
- Skin conductance level (SCL) is typically used to measure the level of electrodermal activity during a period of time
- Because of intrinsic biological differences among participants, baseline values in a resting state before playing the game must be measured and subtracted from in-game measurements

Examples of physiological measurements



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